

Chris H. Kim

professor

基本信息

姓名: Chris H. Kim

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地区: United States

荣誉称号

★ IEEE Fellow

这些成就代表了在学术领域的最高认可，展现了卓越的研究能力和影响力。

学科领域

集成电路设计与集成系统

微电子科学与工程

研究领域

数字集成电路设计

低功耗电路与能量管理

微电子器件与制造

微电子封装与可靠性

EDA工具与电路验证

集成电路设计与制造

超大规模集成电路设计

存储器设计与架构

低功耗设计与能耗优化

个人简介

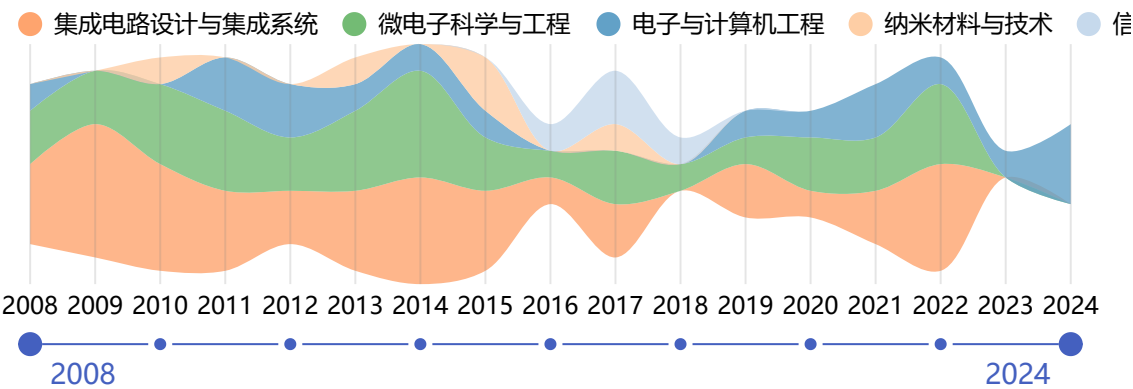
Professor Chris H. Kim, McKnight Presidential Endowed Chair and Distinguished McKnight University Professor at the University of Minnesota, specializes in digital, mixed-signal, and memory IC design. His research focuses on semiconductor reliability, hardware security, and beyond-CMOS technologies. Notably, he has received the Semiconductor Research

Corporation (SRC) Sustainable Future Award, Intel Outstanding Researcher Award, and Taylor Award for Distinguished Research. Professor Kim is an IEEE Fellow and can be reached at chriskim-at-umn-dot-edu.

工作经历

University of Minnesota - Professor - 至今

近年发表的论文



Neuro-Symbolic Computing: Advancements and Challenges in Hardware-Software Co-Design
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS
Yang, Xiaoxuan; Wang, Zhangyang; Hu, X. Sharon; Kim, Chris H.; Yu, Shimeng; Pajic, Miroslav; Manohar, Rajit; Chen, Yiran; Li, Hai Helen
被引用次数: 1 2024

3SAT on an all-to-all-connected CMOS Ising solver chip
SCIENTIFIC REPORTS
Cilasun, Huesrev; Zeng, Ziqing; Ramprasath, S.; Kumar, Abhimanyu; Lo, Hao; Cho, William; Moy, William; Kim, Chris H.; Karpuzcu, Ulya R.; Sapatnekar, Sachin S.
被引用次数: 1 2024

Quantum-Inspired Machine Learning Framework using a Physics-based Ising Solver Chip
2024 IEEE 7TH INTERNATIONAL CONFERENCE ON INDUSTRIAL CYBER-PHYSICAL SYSTEMS, ICPS 2024
Khot, Ameya; Kim, Taesic; Akash, Alve Rahman; Kim, Chris H.; Moy, William
被引用次数: 1 2024

An Ising solver chip based on coupled ring oscillators with a 48-node all-to-all connected array architecture
NATURE ELECTRONICS
Lo, Hao; Moy, William; Yu, Hanzhao; Sapatnekar, Sachin; Kim, Chris H.
被引用次数: 19 2023

Electromigration Assessment in Power Grids with Account of Redundancy and Non-Uniform Temperature Distribution

PROCEEDINGS OF THE 2023 INTERNATIONAL SYMPOSIUM ON PHYSICAL DESIGN, ISPD 2023

Kteyan, Armen; Sukharev, Valeriy; Volkov, Alexander; Choy, Jun-Ho; Najm, Farid N.; Yi, Yong Hyeon; Kim, Chris H.; Moreau, Stephane

被引用次数: 2

2023

A Calibration - Free Synthesizable Odometer Featuring Automatic Frequency Dead Zone Escape and Start - up Glitch Removal

2022 IEEE INTERNATIONAL RELIABILITY PHYSICS SYMPOSIUM (IRPS)

Islam, Tahmida; Kim, Junkyu; Kim, Chris H.; Tipple, David; Nelson, Michael; Jin, Robert; Jarrar, Anis

被引用次数: 0

2022

A Calibration-Free Synthesizable Odometer Featuring Automatic Frequency Dead Zone Escape and Start-up Glitch Removal

2022 IEEE INTERNATIONAL RELIABILITY PHYSICS SYMPOSIUM (IRPS)

Islam, Tahmida; Kim, Junkyu; Kim, Chris H.; Tipple, David; Nelson, Michael; Jin, Robert; Jarrar, Anis

被引用次数: 1

2022

A 16nm All-Digital Hardware Monitor for Evaluating Electromigration Effects in Signal Interconnects Through Bit-Error-Rate Tracking

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Pande, Nakul; Zhou, Chen; Lin, Ming-Hsien; Fung, Rita; Wong, Richard; Wen, Shi-Jie; Kim, Chris H.

被引用次数: 1

2022

A 32Gb/s Time-Based PAM-4 Transceiver for High-Speed DRAM Interfaces With In-Situ Channel Loss and Bit-Error-Rate Monitors

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Chiu, Po-Wei; Kim, Chris H.

被引用次数: 6

2022

An Embedded NAND Flash-Based Compute-In-Memory Array Demonstrated in a Standard Logic Process

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Minsu; Liu, Muqing; Everson, Luke R.; Kim, Chris H.

被引用次数: 21

2022

Experimental Validation of a Novel Methodology for Electromigration Assessment in On-Chip Power Grids

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Sukharev, Valeriy; Kteyan, Armen; Najm, Farid N.; Yi, Yong Hyeon; Kim, Chris H.; Choy, Jun-Ho; Torosyan, Sofya; Zhu, Yu

被引用次数: 3

2022

A 1,968-node coupled ring oscillator circuit for combinatorial optimization problem solving

NATURE ELECTRONICS

Moy, William; Ahmed, Ibrahim; Chiu, Po-wei; Moy, John; Sapatnekar, Sachin S.; Kim, Chris H.

被引用次数: 54

2022

Neutron-Induced Pulsewidth Distribution of Logic Gates Characterized Using a Pulse Shrinking Chain-Based Test Structure

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Pande, Nakul; Kumar, Saurabh; Everson, Luke R.; Park, Gyusung; Ahmed, Ibrahim; Kim, Chris H.

被引用次数: 0

2021

Wide-Range Many-Core SoC Design in Scaled CMOS: Challenges and Opportunities

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Vangal, Sriram; Paul, Somnath; Hsu, Steven; Agarwal, Amit; Kumar, Saurabh; Krishnamurthy, Ram; Krishnamurthy, Harish; Tschanz, James; De, Vivek; Kim, Chris H.

被引用次数: 6

2021

A Back-Sampling Chain Technique for Accelerated Detection, Characterization, and Reconstruction of Radiation-Induced Transient Pulses

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Kumar, Saurabh; Cho, Minki; Everson, Luke R.; Malavasi, Andres; Lake, Dan; Tokunaga, Carlos; Khellah, Muhammad; Tschanz, James W.; De, Vivek; Kim, Chris H.

被引用次数: 1

2021

A Probabilistic Compute Fabric Based on Coupled Ring Oscillators for Solving Combinatorial Optimization Problems

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Ahmed, Ibrahim; Chiu, Po-Wei; Moy, William; Kim, Chris H.

被引用次数: 69

2021

A Time-Based Intra-Memory Computing Graph Processor Featuring A* Wavefront Expansion and 2-D Gradient Control

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Everson, Luke R.; Sapatnekar, Sachin S.; Kim, Chris H.

被引用次数: 7

2021

Extreme Temperature Characterization of Amplifier Response up to 300 Degrees Celsius Using Integrated Heaters and On-chip Samplers

ESSCIRC 2021 - IEEE 47TH EUROPEAN SOLID STATE CIRCUITS CONFERENCE (ESSCIRC)

Yu, Hanzhao; Park, Gyusung; Kim, Chris H.

被引用次数: 1

2021

Understanding the Key Parameter Dependences Influencing the Soft-Error Susceptibility of Standard Combinational Logic

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Pande, Nakul; Kumar, Saurabh; Everson, Luke R.; Kim, Chris H.

被引用次数: 9

2020

Electromigration Effects in Power Grids Characterized From a 65 nm Test Chip

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Zhou, Chen; Fung, Rita; Wen, Shi-Jie; Wong, Richard; Kim, Chris H.

被引用次数: 13

2020

Impact of Process Variability on Write Error Rate and Read Disturbance in STT-MRAM Devices

IEEE TRANSACTIONS ON MAGNETICS

Song, Jeehwan; Dixit, Hemant; Behin-Aein, Behtash; Kim, Chris H.; Taylor, William

被引用次数: 27

2020

Demonstration of a Passive IC Tamper Sensor Based on an Exposed Floating Gate Device in a Standard Logic Process

IEEE TRANSACTIONS ON ELECTRON DEVICES

Liu, Muqing; Kim, Chris H.

被引用次数: 3

2019

Analysis of Neutron-Induced Multibit-Upset Clusters in a 14-nm Flip-Flop Array

IEEE TRANSACTIONS ON NUCLEAR SCIENCE

Kumar, Saurabh; Cho, Minki; Everson, Luke; Tang, Qianying; Meinerzhagen, Pascal; Malavasi, Andres; Lake, Dan; Tokunaga, Carlos; Khellah, Muhammad; Tschanz, James; De, Vivek; Kim, Chris H.

被引用次数: 10

2019

An Energy-Efficient One-Shot Time-Based Neural Network Accelerator Employing Dynamic Threshold Error Correction in 65 nm

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Everson, Luke R.; Liu, Muqing; Pande, Nakul; Kim, Chris H.

被引用次数: 29

2019

A Fully Integrated Digital LDO With Built-In Adaptive Sampling and Active Voltage Positioning Using a Beat-Frequency Quantizer

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kundu, Somnath; Liu, Muqing; Wen, Shi-Jie; Wong, Richard; Kim, Chris H.

被引用次数: 59

2019

A 65-nm 10-Gb/s 10-mm On-Chip Serial Link Featuring a Digital-Intensive Time-Based Decision Feedback Equalizer

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Chiu, Po-Wei; Kundu, Somnath; Tang, Qianying; Kim, Chris H.

被引用次数: 11

2018

Key-Based Dynamic Functional Obfuscation of Integrated Circuits Using Sequentially Triggered Mode-Based Design

IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY

Koteshwara, Sandhya; Kim, Chris H.; Parhi, Keshab K.

被引用次数: 34

2018

A 0.2-1.45-GHz Subsampling Fractional-N Digital MDLL With Zero-Offset Aperture PD-Based Spur Cancellation and In Situ Static Phase Offset Detection

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kundu, Somnath; Kim, Bongjin; Kim, Chris H.

被引用次数: 23

2017

Characterizing the Impact of RTN on Logic and SRAM Operation Using a Dual Ring Oscillator Array Circuit

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Tang, Qianying; Kim, Chris H.

被引用次数: 8

2017

A Circuit-Based Approach for Characterizing High Frequency Electromigration Effects

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Zhou, Chen; Wang, Xiaofei; Fung, Rita; Wen, Shi-Jie; Wong, Richard; Kim, Chris H.

被引用次数: 9

2017

Solution-Processed Carbon Nanotube True Random Number Generator

NANO LETTERS

Rojas, William A. Gaviria; McMorow, Julian J.; Geier, Michael L.; Tang, Qianying; Kim, Chris H.; Marks, Tobin J.; Hersam, Mark C.

被引用次数: 44

2017

Reliable PUF-Based Local Authentication With Self-Correction

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Lao, Yingjie; Yuan, Bo; Kim, Chris H.; Parhi, Keshab K.

被引用次数: 30

2017

Dynamic Memory Cells Using MoS₂ Field-Effect Transistors Demonstrating Femtoampere Leakage Currents

ACS NANO

Kshirsagar, Chaitanya U.; Xu, Weichao; Su, Yang; Robbins, Matthew C.; Kim, Chris H.; Koester, Steven J.

被引用次数: 62

2016

Beat Frequency Detector-Based High-Speed True Random Number Generators: Statistical Modeling and Analysis

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

Lao, Yingjie; Tang, Qianying; Kim, Chris H.; Parhi, Keshab K.

被引用次数: 5

2016

A 0.0054-mm² Frequency-to-Current Conversion-Based Fractional Frequency Synthesizer in 32 nm Utilizing Deep Trench Capacitor

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Kundu, Somnath; Kim, Chris H.

System-Level Power Analysis of a Multicore Multipower Domain Processor With ON-Chip Voltage Regulators

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Paul, Ayan; Park, Sang Phill; Somasekhar, Dinesh; Kim, Young Moon; Borkar, Nitin; Karpuzcu, Ulya R.; Kim, Chris H.

被引用次数: 7

2016

Solution-processed carbon nanotube thin-film complementary static random access memory

NATURE NANOTECHNOLOGY

Geier, Michael L.; McMorrow, Julian J.; Xu, Weichao; Zhu, Jian; Kim, Chris H.; Marks, Tobin J.; Hersam, Mark C.

被引用次数: 198

2015

Large-Area, Low-Voltage, Antiambipolar Heterojunctions from Solution-Processed Semiconductors

NANO LETTERS

Jariwala, Deep; Sangwan, Vinod K.; Seo, Jung-Woo Ted; Xu, Weichao; Smith, Jeremy; Kim, Chris H.; Lauhon, Lincoln J.; Marks, Tobin J.; Hersam, Mark C.

被引用次数: 115

2015

A Ring-Oscillator-Based Reliability Monitor for Isolated Measurement of NBTI and PBTI in High-k/Metal Gate Technology

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Kim, Tony Tae-Hyoung; Lu, Pong-Fei; Jenkins, Keith A.; Kim, Chris H.

被引用次数: 20

2015

The Dependence of BTI and HCI-Induced Frequency Degradation on Interconnect Length and Its Circuit Level Implications

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Wang, Xiaofei; Tang, Qianying; Jain, Pulkit; Jiao, Dong; Kim, Chris H.

被引用次数: 11

2015

Spin-Based Computing: Device Concepts, Current Status, and a Case Study on a High-Performance Microprocessor

PROCEEDINGS OF THE IEEE

Kim, Jongyeon; Paul, Ayan; Crowell, Paul A.; Koester, Steven J.; Sapatnekar, Sachin S.; Wang, Jian-Ping; Kim, Chris H.

被引用次数: 138

2015

Overstress-Free 4 x VDD Switch in a Generic Logic Process Supporting High and Low Voltage Modes

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Song, Seung-Hwan; Kim, Jongyeon; Kim, Chris H.

被引用次数: 1

2015

A Comparative Study of Single-Poly Embedded Flash Memory Disturbance, Program/Erase Speed, Endurance, and Retention Characteristic

IEEE TRANSACTIONS ON ELECTRON DEVICES

Song, Seung-Hwan; Kim, Jongyeon; Kim, Chris H.

被引用次数: 10

2014

Aging Statistics Based on Trapping/Detrapping: Compact Modeling and Silicon Validation

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Sutaria, Ketul B.; Velamala, Jyothi Bhaskarr; Kim, Chris H.; Sato, Takashi; Cao, Yu

被引用次数: 23

2014

Distributed On-Chip Switched-Capacitor DC-DC Converters Supporting DVFS in Multicore Systems

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Zhou, Pingqiang; Paul, Ayan; Kim, Chris H.; Sapatnekar, Sachin S.

被引用次数: 16

2014

Aerosol Jet Printed, Sub-2 V Complementary Circuits Constructed from P- and N-Type Electrolyte Gated Transistors

ADVANCED MATERIALS

Hong, Kihyon; Kim, Yong Hyun; Kim, Se Hyun; Xie, Wei; Xu, Weichao David; Kim, Chris H.; Frisbie, C. Daniel

被引用次数: 97

2014

SILICON ODOMETERS: COMPACT IN SITU AGING SENSORS FOR ROBUST SYSTEM DESIGN

IEEE MICRO

Wang, Xiaofei; Keane, John; Kim, Tony Tae-Hyoung; Jain, Pulkit; Tang, Qianying; Kim, Chris H.

被引用次数: 16

2014

A Bit-by-Bit Re-Writable Eflash in a Generic 65 nm Logic Process for Moderate-Density Nonvolatile Memory Applications

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Song, Seung-Hwan; Chun, Ki Chul; Kim, Chris H.

被引用次数: 15

2014

A Supply-Noise Sensitivity Tracking PLL in 32 nm SOI Featuring a Deep Trench Capacitor Based Loop Filter

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Bongjin; Xu, Weichao; Kim, Chris H.

被引用次数: 7

2014

A Write-Back-Free 2T1D Embedded DRAM With Local Voltage Sensing and a Dual-Row-Access Low Power Mode

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Zhang, Wei; Chun, Ki Chul; Kim, Chris H.

被引用次数: 14

2013

Subnanowatt Carbon Nanotube Complementary Logic Enabled by Threshold Voltage Control

NANO LETTERS

Geier, Michael L.; Prabhumirashi, Pradyumna L.; McMorrow, Julian J.; Xu, Weichao; Seo, Jung-Woo T.; Everaerts, Ken; Kim, Chris H.; Marks, Tobin J.; Hersam, Mark C.

被引用次数: 98

2013

Aerosol Jet Printed, Low Voltage, Electrolyte Gated Carbon Nanotube Ring Oscillators with Sub-5 μ s Stage Delays

NANO LETTERS

Ha, Mingjing; Seo, Jung-Woo T.; Prabhumirashi, Pradyumna L.; Zhang, Wei; Geier, Michael L.; Renn, Michael J.; Kim, Chris H.; Hersam, Mark C.; Frisbie, C. Daniel

被引用次数: 231

2013

Aerosol-Jet-Printed, 1 Volt H-Bridge Drive Circuit on Plastic with Integrated Electrochromic Pixel

ACS APPLIED MATERIALS & INTERFACES

Ha, Mingjing; Zhang, Wei; Braga, Daniele; Renn, Michael J.; Kim, Chris H.; Frisbie, C. Daniel

被引用次数: 47

2013

A Logic-Compatible Embedded Flash Memory for Zero-Standby Power System-on-Chips Featuring a Multi-Story High Voltage Switch and a Selective Refresh Scheme

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Song, Seung-Hwan; Chun, Ki Chul; Kim, Chris H.

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2013

A Scaling Roadmap and Performance Evaluation of In-Plane and Perpendicular MTJ Based STT-MRAMs for High-Density Cache Memory

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Chun, Ki Chul; Zhao, Hui; Harms, Jonathan D.; Kim, Tae-Hyoung; Wang, Jian-Ping; Kim, Chris H.

被引用次数: 336

2013

An SRAM Reliability Test Macro for Fully Automated Statistical Measurements of VMIN Degradation

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Kim, Tony Tae-Hyoung; Zhang, Wei; Kim, Chris H.

被引用次数: 9

2012

Design, Modeling, and Test of a Programmable Adaptive Phase-Shifting PLL for Enhancing Clock Data Compensation

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jiao, Dong; Kim, Bongjin; Kim, Chris H.

被引用次数: 8

2012

A 2T1C Embedded DRAM Macro With No Boosted Supplies Featuring a 7T SRAM Based Repair and a Cell Storage Monitor

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Chun, Ki Chul; Zhang, Wei; Jain, Pulkit; Kim, Chris H.

被引用次数: 38

2012

A 667 MHz Logic-Compatible Embedded DRAM Featuring an Asymmetric 2T Gain Cell for High Speed On-Die Caches

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Chun, Ki Chul; Jain, Pulkit; Kim, Tae-Ho; Kim, Chris H.

被引用次数: 178

2012

Adaptive Techniques for Overcoming Performance Degradation Due to Aging in CMOS Circuits

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Kumar, Sanjay V.; Kim, Chris H.; Sapatnekar, Sachin S.

被引用次数: 59

2011

AN ODOMETER FOR CPUS

IEEE SPECTRUM

Keane, John; Kim, Chris H.

被引用次数: 35

2011

An Array-Based Test Circuit for Fully Automated Gate Dielectric Breakdown Characterization

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Keane, John; Venkatraman, Shrinivas; Butzen, Paulo; Kim, Chris H.

被引用次数: 6

2011

A 3T Gain Cell Embedded DRAM Utilizing Preferential Boosting for High Density and Low Power On-Die Caches

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Chun, Ki Chul; Jain, Pulkit; Lee, Jung Hwa; Kim, Chris H.

被引用次数: 116

2011

An Array-Based Odometer System for Statistically Significant Circuit Aging Characterization

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Keane, John; Zhang, Wei; Kim, Chris H.

被引用次数: 30

2011

On-chip reliability monitors for measuring circuit degradation

MICROELECTRONICS RELIABILITY

Keane, John; Kim, Tae-hyoung; Wang, Xiaofei; Kim, Chris H.

被引用次数: 22

2010

Printed, Sub-3V Digital Circuits on Plastic from Aqueous Carbon Nanotube Inks

ACS NANO

Ha, Mingjing; Xia, Yu; Green, Alexander A.; Zhang, Wei; Renn, Mike J.; Kim, Chris H.; Hersam, Mark C.; Frisbie, C. Daniel

被引用次数: 398

2010

Printed Sub-2 V Gel-Electrolyte-Gated Polymer Transistors and Circuits

ADVANCED FUNCTIONAL MATERIALS

Xia, Yu; Zhang, Wei; Ha, Mingjing; Cho, Jeong Ho; Renn, Michael J.; Kim, Chris H.; Frisbie, C. Daniel

被引用次数: 195

2010

An All-In-One Silicon Odometer for Separately Monitoring HCI, BTI, and TDDB

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Keane, John; Wang, Xiaofei; Persaud, Devin; Kim, Chris H.

被引用次数: 147

2010

An On-Chip NBTI Sensor for Measuring pMOS Threshold Voltage Degradation

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Keane, John; Kim, Tae-Hyoung; Kim, Chris H.

被引用次数: 63

2010

Circuit Design and Modeling Techniques for Enhancing the Clock-Data Compensation Effect Under Resonant Supply Noise

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Jiao, Dong; Gu, Jie; Kim, Chris H.

被引用次数: 21

2010

A Finite-Oxide Thickness-Based Analytical Model for Negative Bias Temperature Instability

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Kumar, Sanjay V.; Kim, Chris H.; Sapatnekar, Sachin S.

被引用次数: 31

2009

Sleep Transistor Sizing and Adaptive Control for Supply Noise Minimization Considering Resonance

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Gu, Jie; Eom, Hanyong; Keane, John; Kim, Chris H.

被引用次数: 8

2009

On-Chip Supply Noise Regulation Using a Low-Power Digital Switched Decoupling Capacitor Circuit

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Gu, Jie; Eom, Hanyong; Kim, Chris H.

被引用次数: 25

2009

A Voltage Scalable 0.26 V, 64 kb 8T SRAM With Vmin Lowering Techniques and Deep Sleep Mode

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Tae-Hyoung; Liu, Jason; Kim, Chris H.

被引用次数: 106

2009

Design and Implementation of Active Decoupling Capacitor Circuits for Power Supply Regulation in Digital ICs

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Gu, Jie; Harjani, Ramesh; Kim, Chris H.

被引用次数: 29

2009

Body bias voltage computations for process and temperature compensation

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Kumar, Sanjay V.; Kim, Chris H.; Sapatnekar, Sachin S.

被引用次数: 43

2008

A high-speed variation-tolerant interconnect technique for sub-threshold circuits using capacitive boosting

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Kil, Jonggab; Gu, Jie; Kim, Chris H.

被引用次数: 35

2008

Silicon odometer: An on-chip reliability monitor for measuring frequency degradation of digital circuits

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Tae-Hyoung; Persaud, Randy; Kim, Chris H.

被引用次数: 216

2008

A 0.2 V, 480 kb subthreshold SRAM with 1 k cells per bitline for ultra-low-voltage computing

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kim, Tae-Hyoung; Liu, Jason; Keane, John; Kim, Chris H.

被引用次数: 168

2008

Statistical leakage estimation of double gate FinFET devices considering the width quantization property

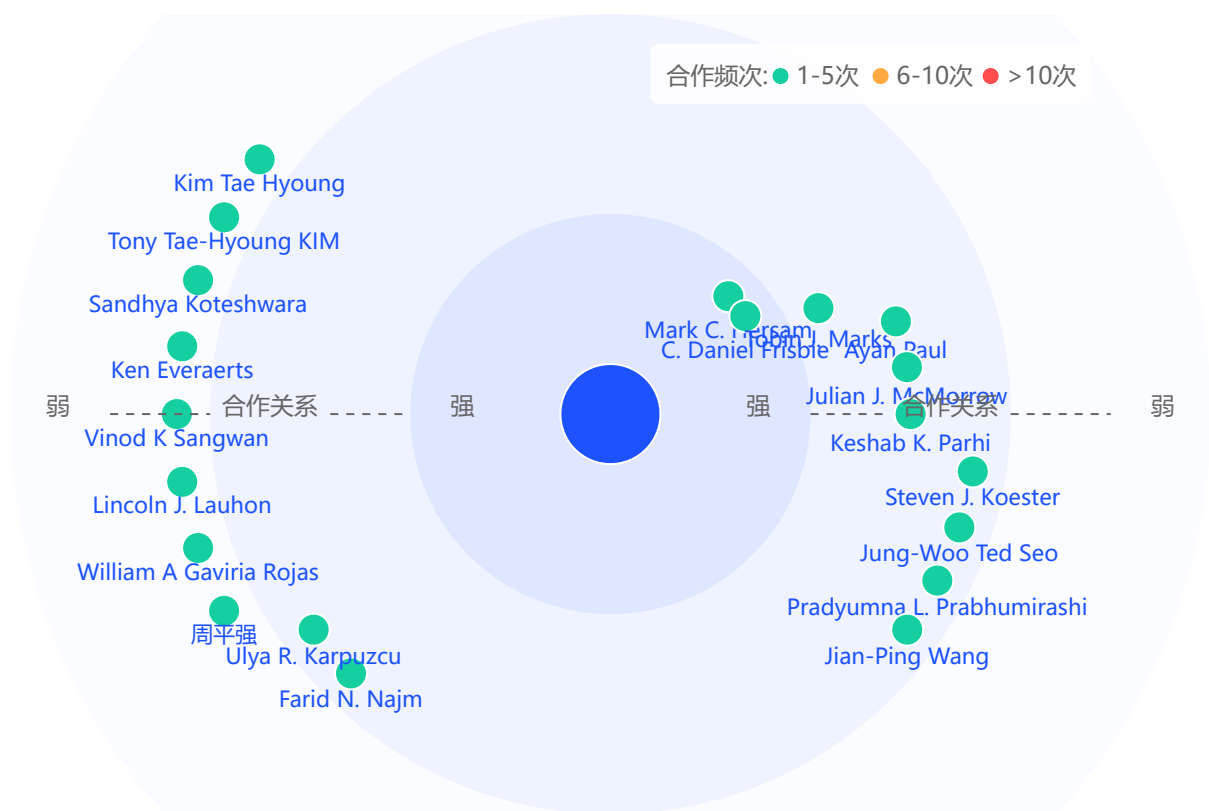
IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Gu, He; Keane, John; Sapatnekar, Sachin; Kim, Chris H.

被引用次数: 18

2008

学术合作



Mark C. Hersam

西北大学(埃文斯顿)

合作次数: 5

C. Daniel Frisbie

明尼苏达大学-双城

合作次数: 5

Tobin J. Marks

西北大学(埃文斯顿)

合作次数: 4

Ayan Paul

明尼苏达大学-双城

合作次数: 3

Julian J. McMorow

西北大学(埃文斯顿)

合作次数: 3

还有 20 个合作者